Working Notes on Calabi–Yau Quantum Groups

Raeez Lorgat

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Contents

1 Overview and programme

These working notes record the development of Volume III (Calabi–Yau Quantum Groups) of the modular Koszul duality programme. The central thesis: *the combinatorics of a CY3 threefold X constitute the generalized root datum of a quantum vertex chiral group* $G(X)$. The automorphic correction of the underlying BKM superalgebra is the shadow Postnikov tower Θ_A from Volume I.

1.1 The four target theorems

- **CY-A** (CY-to-chiral functor): $\Phi\colon \mathrm{CY}_d\text{-Cat} \to \mathsf{E}_2\text{-ChirAlg}$.

- **CY-B** (E_2-chiral Koszul duality): Automorphic correction = shadow tower; denominator identity = bar Euler product.

- **CY-C** (Quantum group realization): $\mathrm{Rep}^{\mathsf{E}_2}(G(X))$ is braided monoidal; for toric CY3, this gives the affine super Yangian.
- **CY-D** (Modular CY characteristic): $\kappa(G(X))$ = weight of the automorphic form.

1.2 Current status of the programme

Component	Status	Notes file
Generalized root datum axioms	Draft (CY1–CY7)	theory_generalized_root_datum
E_2-chiral algebra formalism	Draft (1243 lines)	theory_e2_chiral_formalism
Automorphic = shadow tower	Draft (Thm + proof)	theory_automorphic_shadow
CY-to-chiral functor	Draft (4-step construction)	theory_cy_to_chiral_construction
$\mathrm{CY}_2 \to \mathrm{CY}_3$ fibration	Draft (Borcherds lift)	theory_cy2_cy3_fibration
Denominator = bar Euler product	Draft (3-stage proof)	theory_denominator_bar_euler
QVCG Koszul duality	Draft (Langlands conj)	theory_qvcg_koszul
Drinfeld = chiral center	Draft (factorization proof)	theory_drinfeld_chiral_center
CoHA = E_1-sector	Draft (Drinfeld double)	theory_coha_e1_sector
KL in E_2-chiral framework	Draft (CY category conj)	theory_kl_e2_chiral
Higgs sheaves as CY_2 QVCGs	Draft (genus filtration)	theory_higgs_cy2_qvcg
BPS = root multiplicities	Draft	physics_bps_root_multiplicities
Topological strings & root data	Draft (BCOV = MC)	physics_topological_strings
M-theory branes / holography	Draft	physics_mtheory_branes
4d $\mathcal{N} = 2$ / Hitchin	Draft (Z_{Nek} conj)	physics_4d_n2_hitchin
Wall-crossing = MC gauge equiv	Draft	physics_wall_crossing_mc
S-duality = Langlands = Koszul	Draft	physics_sduality_langlands
Anomaly cancellation / κ	Draft ($\kappa \neq \chi/12$)	physics_anomaly_cancellation
BV-BRST for CY categories	Draft (QME = MC)	physics_bv_brst_cy
3d mirror symmetry / CY_2	Draft	physics_3d_mirror
Celestial holography / CY QG	Draft ($\mathcal{W}_{1+\infty} = G(\mathbb{C}^3)$)	physics_celestial_cy
Hitchin / geometric Langlands	Draft	physics_hitchin_langlands
$\phi_{0,1}$ Fourier coefficients	Computed (51 tests)	compute/lib/phi01_fourier
$\mathbb{C}^3$ DT / MacMahon	Computed (57 tests)	compute/lib/c3_dt_partition
Lattice data (8 dd-forms)	Computed (65 tests)	compute/lib/dd_modular_lattices
Topological vertex	Computed	compute/lib/topological_vertex
Igusa product formula	Computed	compute/lib/igusa_product_formula
Affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$	Computed (81 tests)	compute/lib/affine_yangian_gl1
BKM shadow tower	Computed (67 tests)	compute/lib/bkm_shadow_tower
Elliptic Hall algebra	Computed (78 tests)	compute/lib/elliptic_hall
CY Euler characteristics	Computed (85 tests)	compute/lib/cy_euler
WKB denominator	Computed	compute/lib/wkb_denominator
Higgs CoHA on $\mathbb{P}^1$	Computed (84 tests)	compute/lib/higgs_p1_coha

2 Key identifications

2.1 The master dictionary

CY3 geometry	BKM / Yangian	Vol I bar-cobar
Lattice $\Lambda(X)$	Root lattice	Cyclic deformation complex
BPS invariants DT_α	Root multiplicities $\mathrm{mult}(\alpha)$	Shadow tower components
Toric diagram	Dynkin diagram / Gram matrix	Collision r-matrix $r(z)$
Automorphic correction	Imaginary roots added to $\mathfrak{g}$	Higher-arity shadows
Denominator identity	WKB character formula	Bar Euler product
$\mathrm{Sp}_4(\mathbb{Z})$ / Weyl group	Symmetry group	E_2 braiding
$\phi_{0,1}$ (K3 elliptic genus)	Root multiplicity function	Shadow depth classification
CoHA (positive half)	$U(\mathfrak{n}_+)$	E_1-chiral sector
Full Yangian / BKM	Full algebra $\mathfrak{g}_X$	E_2-chiral algebra $A(X)$

2.2 The central identification

Theorem 2.1 *(Automorphic correction = shadow tower).* The shadow Postnikov tower $\Theta^{\leq r}_{A_X}$ of the quantum vertex chiral group $G(X)$ captures exactly the roots of complexity $\leq r$ in the BKM root system:

(a) Arity 2 = real roots + Weyl vector (modular characteristic κ).

(b) Arity 3 = first layer of imaginary roots (cubic shadow C).

(c) Arity r = roots up to depth $r-2$ in the positive cone.

(d) The denominator identity of $\mathfrak{g}_X$ equals the bar-complex Euler product of $B(A_X)$.

2.3 Computationally verified

For $K3 \times E$ with $\mathfrak{g}_{\Delta_5}$:

- Arity 2: 21 real roots (norm 2), all multiplicity 2. $\kappa = 5$.
- Arity 3: 2 lightlike imaginary roots at $(1,0,0)$ and $(0,0,1)$, multiplicity 20.
- Arity 4: 7 roots including the timelike root $(1,0,1)$ with multiplicity -252.
- At every truncation order $r = 2,\dots,6$: product formula matches shadow projection.

3 The $K3 \times E$ example

3.1 The geometry

$X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, S = K3, E = elliptic curve. DT partition function $Z^X = C/(\Delta_5)^2$.

3.2 The lattice

$\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$, signature $(3,2)$. Hyperbolic sublattice $\Lambda^{2,1}_{II}$ with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Weyl vector $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$, $(\rho, \delta_i) = -1$. $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$.

3.3 The modular characteristic

$\kappa(G(K3 \times E)) = 5 = \mathrm{weight}(\Delta_5) = h^{1,1}(K3)/4 = (\chi(K3) - 4)/4$.

This is NOT $\chi(X)/12$ (which gives 0, since $\chi(K3 \times E) = 0$). The modular characteristic counts the worldsheet anomaly (full BPS spectrum), not the target-space anomaly (massless modes).

4 The toric CY3 / CoHA example

For $\mathbb{C}^3$: CoHA $= Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$ at the self-dual level. $\dim Y^+_n = p(n)$ (plane partitions). The MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the denominator identity. The affine Yangian structure function $g(z) = (z-h_1)(z-h_2)(z-h_3)/((z+h_1)(z+h_2)(z+h_3))$ with $h_1 + h_2 + h_3 = 0$. Even ϕ-coefficients vanish: $\phi_2 = \phi_4 = 0$ because $\log g$ has only odd powers.

5 Higgs sheaves and the CY_2 escape

The most natural escape from CY3: $\mathrm{Higgs}(C) = \mathrm{Coh}(T^*C)$ is CY2. The $\mathbb{S}^2$-framing gives E_2-structure directly.

- **Genus 0** ($C = \mathbb{P}^1$): Yangian $Y(\mathfrak{gl}_2)$. Rational R-matrix.
- **Genus 1** ($C = E$): Elliptic Hall algebra $E_{q,t}$ = spherical DAHA. Elliptic R-matrix.
- **Genus** ≥ 2: New "Hitchin Hall algebras." R-matrices on $C \times C$.

Key principle: $\mathrm{CY}_3 = \mathrm{CY}_2$ + automorphic correction (Borcherds lift).

6 The Langlands connection

Conjecture 6.1 (Langlands = Koszul). For a curve C and reductive group G, the quantum vertex chiral group of the Hitchin system satisfies

$$G(C, G)^! = G(C, {}^L G).$$

Langlands duality is Koszul duality of quantum vertex chiral groups.

Evidence: Feigin–Frenkel center at critical level, root/coroot exchange, SYZ mirror symmetry of $\mathcal{M}_H(C, G)$ and $\mathcal{M}_H(C, {}^L G)$, Kapustin–Witten.

7 Open questions

1. What is the generalized root datum of the quintic? (Non-toric, no quiver.)
2. Is the chain-level $\mathbb{S}^3$-framing for CY3 categories constructible? (Gap in CY-A for $d = 3$.)
3. What is the explicit automorphic form (denominator identity) for Hitchin systems with $G = \mathrm{SL}_3$ or E_8?
4. How does Kontsevich–Soibelman wall-crossing manifest as gauge equivalence of MC elements? (Conjectured, not proved.)
5. Is the formula $\kappa = h^{1,1}/4$ universal for K3-fibered CY3, or specific to $K3 \times E$?
6. For toric CY3 with compact 4-cycles, what replaces the RSYZ theorem?
7. The eight diagonal divisor Siegel modular forms: are they all denominator identities?
8. How does the shadow depth classification (G/L/C/M) manifest for quantum vertex chiral groups?
9. Explicit construction of the E_2-bar complex $B_{\mathsf{E}_2}(A)$ for $A = V_k(\mathfrak{g})$.
10. Quantum geometric Langlands as E_2-chiral Koszul duality: $G(C, \mathfrak{g}, k)^! = G(C, {}^L\mathfrak{g}, k^L)$.

8 Arithmetic shadows and number theory

The shadow tower of a chiral algebra carries arithmetic content that connects to classical number theory through three channels: modular forms at genus 1, Siegel modular forms at genus 2, and multiple zeta values at genus 0.

8.1 MZV periods from the genus-0 bar complex

The genus-0 bar amplitudes on $\mathcal{M}_{0,n}$ produce MZV periods via iterated integrals of the bar propagator $d\log(z_i - z_j)$. The shadow depth classification controls which MZV weights appear.

THEOREM 8.1 *(Shadow–MZV dictionary* (proved in Vol I)*).* For a chirally Koszul algebra $\mathcal{A}$:

(i) $\kappa(\mathcal{A})$ controls the coefficient of $\zeta(2)$ in genus-0 four-point amplitudes.

(ii) The cubic shadow $S_3(\mathcal{A})$ controls the coefficient of $\zeta(3)$ in five-point amplitudes.

(iii) The quartic shadow $S_4(\mathcal{A})$ and contact class Q^{contact} control weight-4 MZV content.

(iv) At general weight r: the arity-r shadow S_r contributes to the MZV space of dimension $d_r = d_{r-2}+d_{r-3}$ (Brown 2012).

Class **G** algebras see only $\zeta(2)$; class **L** see $\zeta(2), \zeta(3)$; class **C** see $\zeta(2), \zeta(4)$ ($S_3 = 0$ but $S_4 \neq 0$); class **M** see MZVs at all weights.

For quantum vertex chiral groups, this dictionary extends to the genus-0 bar complex on a curve C: the MZV periods are replaced by periods of $\mathcal{M}_{0,n}(C)$, which include elliptic MZVs (for $C = E$ an elliptic curve) and higher-genus analogues.

Conjecture 8.2 *(CY MZV dictionary).* For the quantum vertex chiral group $G(C, \mathfrak{g})$, the genus-0 bar amplitudes on $\mathcal{M}_{0,n}(C)$ produce periods of the motivic cohomology of C. The shadow depth of $G(C, \mathfrak{g})$ controls the motivic weight filtration level of these periods.

8.2 The Drinfeld associator as bar transport

The Drinfeld associator $\Phi(A, B)$ is the parallel transport of the KZ connection on $\mathcal{M}_{0,4}$, identified with the genus-0 arity-2 shadow connection. Its MZV coefficients are:

$$\log\Phi = \zeta(2)\,[A,B] + \zeta(3)\,([A,[A,B]] - [B,[A,B]]) \\ + \tfrac{\zeta(4)}{4}\,([A,[A,[A,B]]] + [B,[B,[A,B]]]) + \cdots$$

For quantum vertex chiral groups: the Drinfeld associator governs the genus-0 transport between different trivializations of the bar complex, and the associator's MZV coefficients are the perturbative data of the quantum group structure.

8.3 Siegel modular forms from genus-2 amplitudes

At genus 2, the shadow tower's scalar projection determines the universal part of the genus-2 amplitude, but the full genus-2 partition function carries arithmetic content *beyond* the shadow tower. For lattice VOAs of rank d, the genus-2 partition function is the Siegel theta series $\Theta^{(2)}_\Lambda(\Omega) \in M_{d/2}((4,))$; the shadow tower fixes $F_2 = \kappa\cdot\lambda_2^{\mathrm{FP}}$ but not the Siegel cusp component (Vol I, lattice chapter).

For the quantum vertex chiral group programme: the genus-2 amplitude of $G(C, \mathfrak{g})$ should produce a Siegel modular form on $(4,)$ whose Fourier coefficients encode the BPS spectrum. The Böcherer factorization converts these Fourier coefficients to central L-values, providing algebraic access to the critical line $\Re(s) = 1/2$.

Conjecture 8.3 *(Genus-2 BPS = Böcherer).* For the quantum vertex chiral group $G(K3\times E, \mathfrak{g})$ at the lattice VOA point: the genus-2 MC equation determines the Siegel theta series, and the Böcherer coefficient extracts the central L-value $L(1/2, \pi_f \times \chi_D)$ from the MC data.

8.4 Moonshine and Niemeier atlas

The 24 Niemeier lattices (even unimodular in dimension 24) all have $\kappa = 24$, class $\mathbf{G}$, shadow depth 2. The shadow tower is blind to their root systems: all 24 produce identical shadow data. Distinguishing them requires the full theta series $\Theta_\Lambda = E_{12} + c_\Delta\cdot\Delta$ where $c_\Delta = (691|R| - 65520)/691$.

The Monster module $V^\natural$ has $\kappa = 12$ (from the Virasoro sector alone, since $\dim V_1 = 0$). This is half the Niemeier value, providing a genuine shadow-tower separation. The Monster module is conjecturally class $\mathbf{M}$ (infinite shadow depth), reflecting the nonlinear Griess algebra structure.

For the CY quantum group programme: the K3 quantum vertex chiral group should encode the Mathieu moonshine (umbral moonshine for $\Lambda = 24A_1, G = M_{24}$) through the genus-1 shadow tower. The mock modular forms of umbral moonshine may arise as the quasi-modular shadow content (through the E_2^* propagator dependence).